

Aggregations-Basic comands

```
> db.students.aggregate([
  { $match: { marks: { $gt: 80 } } }
])
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 92,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
```

>_mongo>

```
> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      totalStudents: { $sum: 1 }
    }
  }
])
< {
  _id: 'AI&DS',
  totalStudents: 2
}
{
  _id: 'ECE',
  totalStudents: 2
}
{
  _id: 'CSE',
  totalStudents: 5
}
{
  _id: 'IT',
  totalStudents: 3
}
{
  _id: null,
  totalStudents: 1
}
> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" }
    }
  }
])
```

```

> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" }
    }
  }
])
< {
  _id: 'AI&DS',
  averageMarks: 90
}
{
  _id: 'ECE',
  averageMarks: 73.5
}
{
  _id: 'CSE',
  averageMarks: 90.8
}
{
  _id: 'IT',
  averageMarks: 84.66666666666667
}
{
  _id: null,
  averageMarks: null
}

```

```

> db.students.aggregate([
  {
    $group: {
      _id: null,
      highestMarks: { $max: "$marks" }
    }
  }
])
< {
  _id: null,
  highestMarks: 95
}

```

```

> // Total marks
db.students.aggregate([
  {
    $group: {
      _id: null,
      totalMarks: { $sum: "$marks" }
    }
  }
])
< {
  _id: null,
  totalMarks: 1035
}

```

Aggregations with Multiple Stages

```

}
> db.students.aggregate([
  { $match: { department: "CSE" } },
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" },
      totalStudents: { $sum: 1 }
    }
  },
  { $sort: { averageMarks: -1 } }
])
< {
  _id: 'CSE',
  averageMarks: 90.8,
  totalStudents: 5
}

```

```
> db.students.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 },
  {
    $project: {
      _id: 0,
      rollNo: 1,
      name: 1,
      marks: 1
    }
  }
])
< {
  rollNo: 105,
  name: 'Kiran',
  marks: 95
}
{
  rollNo: 110,
  name: 'Meena',
  marks: 93
}
{
  rollNo: 101,
  name: 'John',
  marks: 92
}
studentDB>
```